

BYPASS GUIDE FOR STARCOOL CIM-6 /6.1 controller

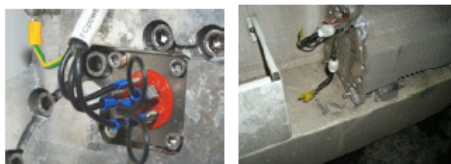
Controller defective

If the controller of the unit is malfunctioning or defective and no replacement is available, the unit can be run in a emergency mode with deteriorated temperature controlling performance. This operation is only recommended for cargo being transported in frozen mode, setpoint below -10°C (14°F). If the cargo is transported in chilled mode, it is recommended that the defective controller is replaced with one from a unit operating in frozen mode.

The emergency procedure for a unit with a defective controller, running in frozen mode, is as follows:

1. Dismount the frequency converter and connect the compressor directly to the power supply. Use the contactor in the controller as a main switch for the compressor.
2. Connect the evaporator fans in low speed and the condenser fan in high speed directly to the power supply.
3. Mount permanent magnet on the electronical expansion valve, pos. 18 (P & I diagram).
4. Install a service gauge manifold on the unit to the two evacuation points, pos. 6 & 27 (P & I – diagram).
5. Close the stop valve pos. 14 (P & I – diagram).
6. Energize the fans and the compressor.
7. Carefully open the stopvalve pos. 14 (P & I – diagram), observing the pressure readings on the service gauge manifold as this is done.
8. For a setpoint temperature of -20°C (-4,0°F) a suction pressure of -24°C (-11,2°F) is to be maintained. In general the suction pressure (temperature) is to be 5°C (7°F) below the set point temperature. In general the discharge pressure (temperature) is to be 10°C (18°F) above ambient temperature.
9. If there is ice building up on the compressor, it indicates that there is coming to much liquid

FC bypass



First, defective FC to be dismantled and 3 phases cables at FC to be directly connected to the compressor terminals, please see attached photo. There are two cables in FC (Frequency Converter). One is the FC power cable as attached photo and other one is FC communication cable. Communication cable to be insulated by using insulation tape.

Also, a wire-jumper has to be fitted on the remaining 3 terminals.

After wiring connections have been done, please access controller, press "service menu", select F03--> FC TYPE, the parameter as "NONE " to be selected. Then, switch on the power to restart the unit.

Opening of solenoid valve and adjust Freon flow from valve , pos:14

Opening of Vexp
(expension valve) by
using emergency
magnet or
permanent magnet

